NTNU | Norwegian University of Science and Technology

Embedded AI

Real-Time Fastener Detection on Raspberry Pi

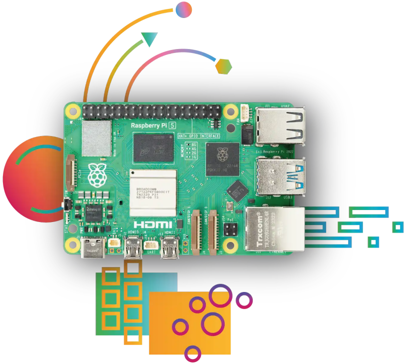
Bukueva Elena

October 27, 2025

Contents

Project goals

- Deploy real-time fastener detection on Raspberry Pi 4 & 5
- Explore edge-AI constraints (latency, power, memory)
- Evaluate state-of-the-art models and embedded performance



Workflow

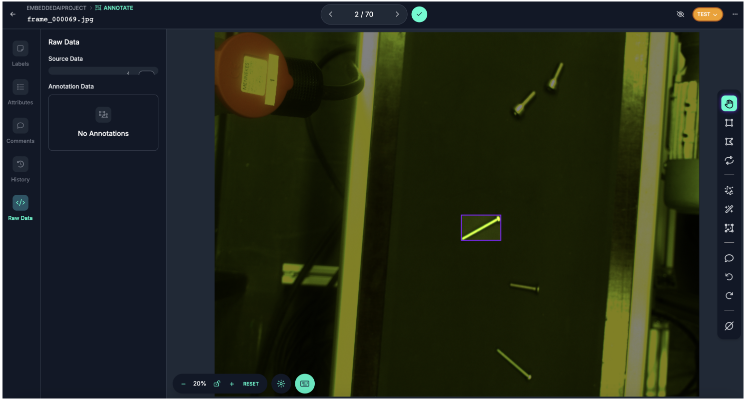
- Set up Raspberry Pi 4/5:
  - Install OS
  - Install drivers for the camera - IDS SDK
- Record videos via the IDS camera, connected to the Raspberry Pi
- Annotate video frames via Roboflow framework
- Run training on the GPU
- Test real-time detection on Raspberry Pi

Camera (UI-3590CP-C-HQ Rev.2)

Sensor	
Resolution	18.10 Mpix (4912×3684)
Gain (master/RGB)	16× / 4×
Model	
Pixel clock range	96–432 MHz
Max frame rate (freerun/trigger)	21 / 21 fps
Exposure time (min–max)	0.013–998 ms
Power consumption	1.5–3.3 W
Image memory	128 MB
Environment & I/O	
Interface	USB 3.0 micro-B, screwable

Roboflow

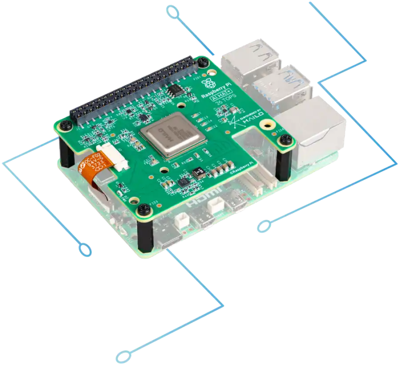
For annotation we used Roboflow. Roboflow is a platform for rapid image annotation, with APIs to upload images and export datasets in model-ready formats.



Raspberry Pi 5

Compute	
SoC	2.4 GHz quad-core 64-bit Arm Cortex-A76 (crypto ext.);
	512 KB L2 per core,
	2 MB shared L3
GPU	VideoCore VII; OpenGL ES 3.1, Vulkan 1.3
Memory and Storage	
RAM	8 GB
microSD	64 GB

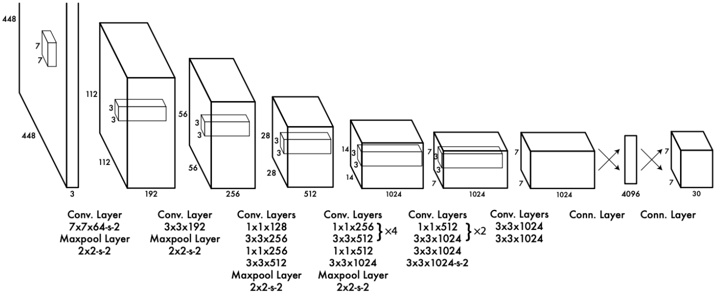
Hailo-8 accelerator offering 26 TOPS inferencing performance respectively



Training

YOLO remains the go-to family for the best speed–accuracy trade-off in object detection. We use the lightweight Small and Nano variants of YOLOv8 (2023) and YOLOv11 (2024) as baselines.

After YOLOv4 the naming became community-driven different teams publish their own "vX" lines. For example, Ultralytics maintains v5, v8, and v11;



Dataset

2131 Total Images

View All Images →

Dataset Split

TRAIN SET 93%	VALID SET 5%	TEST SET 2%
1983 Images	**98** Images	**50** Images

Preprocessing

Auto-Orient: `Applied`
Resize: `Fit (black edges) in 960x960`
Auto-Adjust Contrast: `Using Contrast Stretching`
Grayscale: `Applied`

Augmentations

Outputs per training example: `3`
Flip: `Horizontal, Vertical`
90° Rotate: `Clockwise, Counter-Clockwise, Upside Down`
Hue: `Between -24° and +24°`
Saturation: `Between -25% and +25%`
Brightness: `Between -22% and +22%`
Exposure: `Between -10% and +10%`
Blur: `Up to 2.5px`
Noise: `Up to 1.72% of pixels`

Detection Metrics

For a predicted box p and ground-truth box g:

$$\mathrm{IoU}(p,g) = \frac{|p \cap g|}{|p \cup g|}, \mathrm{P} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \mathrm{R} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \mathrm{AP}_\tau = \int_0^1 p(r;\tau)\,dr,$$

$$\mathrm{mAP}_{0.50} = \frac{1}{C}\sum_{c=1}^{C} \mathrm{AP}_{c,\,0.50},$$

$$\mathrm{mAP}_{[0.50:0.95]} = \frac{1}{C}\sum_{c=1}^{C} \frac{1}{10}\sum_{k=0}^{9} \mathrm{AP}_{c,\,(0.50+0.05k)}.$$

- τ: IoU threshold for TP/FP (0.50 for mAP50; 0.50–0.95 for COCO mAP).
- AP is area under the precision–recall curve (confidence threshold is swept).
- C is the number of object classes we're evaluating over.

Results

(a) YOLO8n

Class	Images	Instances	Box(P)	R	mAP_{50}	mAP_{50-95}
all	98	363	0.927	0.951	0.964	0.733
machine_screw	25	123	0.931	0.967	0.989	0.625
miscellaneous	14	16	0.754	0.812	0.845	0.562
nut	16	41	0.988	1.000	0.995	0.777
washer	20	43	1.000	0.991	0.995	0.906
wood_screw	37	140	0.960	0.986	0.994	0.794

Speed per image (GPU): preprocess 4.5 ms; inference 6.0 ms; loss 0.0 ms; postprocess 2.5 ms.

Speed per image (Raspberry Pi 5): preprocess 8.3 ms; inference 1018.4 ms; loss 0.0 ms; postprocess 1.2 ms.

Speed per image (Raspberry Pi 5 + Hailo): Frames: 50, time: 18.5s, avg FPS: 2.71

Model summary (fused): 72 layers, 3,006,623 parameters, 0 gradients, 8.1 GFLOPs

Results

Table: Validation summaries.

(a) YOLO11s

Class	Images	Instances	Box(P)	R	mAP_{50}	mAP_{50-95}
all	98	363	0.939	0.934	0.966	0.746
machine_screw	25	123	0.877	0.983	0.978	0.649
miscellaneous	14	16	0.899	0.688	0.867	0.565
nut	16	41	0.970	1.000	0.995	0.805
washer	20	43	0.973	1.000	0.995	0.909
wood_screw	37	140	0.978	1.000	0.995	0.802

Speed per image (GPU): preprocess 4.5 ms; inference 9.8 ms; loss 0.0 ms; postprocess 3.7 ms.

Speed per image (Raspberry Pi 5): preprocess 8.4 ms; inference 2728.8 ms; loss 0.0 ms; postprocess 0.9 ms.

YOLO11s summary (fused): 100 layers, 9,414,735 parameters, 0 gradients, 21.3 GFLOPs

Results

Table: Validation summaries.

(a) YOLO11n

Class	Images	Instances	Box(P)	R	mAP_{50}	mAP_{50-95}
all	98	363	0.921	0.935	0.964	0.739
machine_screw	25	123	0.844	0.935	0.957	0.611
miscellaneous	14	16	0.922	0.741	0.888	0.608
nut	16	41	0.970	1.000	0.995	0.799
washer	20	43	0.996	1.000	0.995	0.902
wood_screw	37	140	0.873	1.000	0.983	0.773

Speed per image (GPU): preprocess 0.5 ms; inference 5.6 ms; loss 0.0 ms; postprocess 0.9 ms.

Speed per image (Raspberry Pi 5): preprocess 8.2 ms; inference 998.4 ms; loss 0.0 ms; postprocess 1.0 ms.

100 layers, 2,583,127 parameters, 0 gradients, 6.3 GFLOPs.

Results

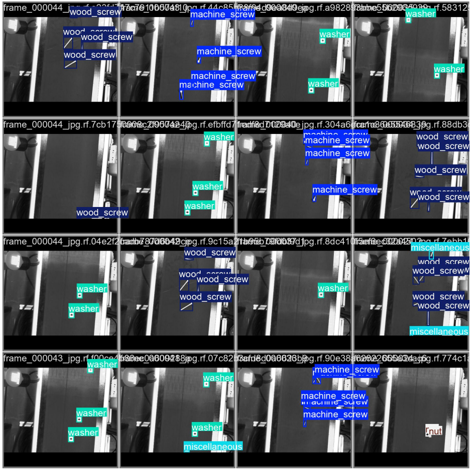


Figure: Ground truth: yolov8n

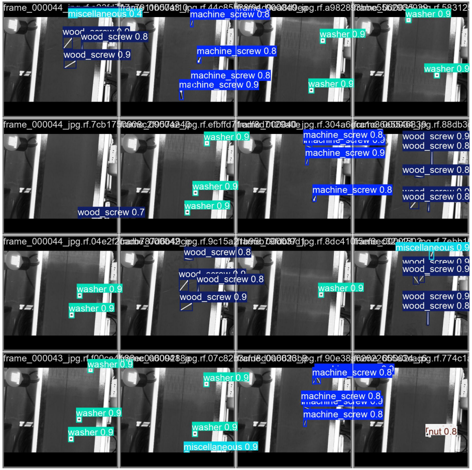


Figure: Prediction: yolov8n

Results

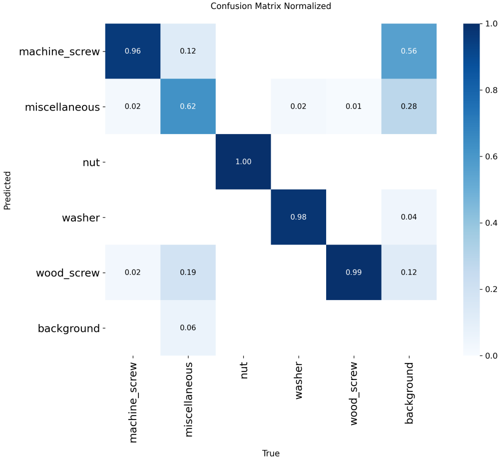


Figure: Normalized Confusion Matrix: yolov8n

References

- Slide 3 (picture): https://www.raspberrypi.com/products/raspberry-pi-5/
- Slide 5 (picture & specifications): https://en.ids-imaging.com/download-details/AB00606.html
- Slide 6 (info & color palette): https://roboflow.com/brand#colors
- Slide 7 (picture): https://www.raspberrypi.com/products/ai-hat/

References

- Redmon, Joseph, et al. "You only look once: Unified, real-time object detection." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.
- Lin, Tsung-Yi, et al. "Microsoft coco: Common objects in context." *European conference on computer vision. Cham: Springer International Publishing*. 2014.